MOUNTAIN BIKE

Ride for everyone …

There are different types of Mtb Trail Difficulty Rating System out there that are designed to answer the question “how difficult is this trail / tour?“
The first thing to know is that an internationally recognized mtb trail difficulty rating system does not exist.

For our mountain bike tours, we adopt the Single Trail Scale from the Gran Canaria, which classifies the difficulty of the trails into 3 classes and 6 levels: S0, S1, S2, S3, S4, S5.

https://grancanariamountainbike.com/mtb-trail-difficulty-rating-system/

The Single Trail Scale (STS) Difficulty Classes

The STS mtb trail difficulty rating system is divided into three main classes of difficulty: easy, medium and difficult, which are identified by the usual color coding used for the ski slopes (**easy: blue**, **medium: red**, **difficult: black**).

These difficulty classes are based on the riding skills of an average mountain biker, equipped with a technically current mountain bike.

The Single Trail Scale (STS) Difficulty Levels

For a concrete classification and accurate differentiation of the trails and difficulty classes, the STS Single Trail Scale also adopts 6 relatively well demarcated difficulty levels (S-grades) from S0 to S5. Thus, the lower end of the scale for an average biker can be taken as **WITHOUT DIFFICULTIES**, while the upper end can be equated to **UNRIDEABLE**.

The difficulty class EASY comprises the degrees **S0** and **S1**, MEDIUM is defined by **S2** and DIFFICULT includes **S3**, **S4** and **S5** grade.

The STS Single Track Scale only assesses the technical difficulty of mountain bike trails. Length and elevation gain / loss are not taken into account. Furthermore, this mtb trail difficulty rating system is independent of elements that do not relate to the mountain bike technique, or variable factors such as:

- trail conditions (mud, ice, slippery rock, etc.),
- danger (risk of falling),
- weather conditions (rain, wind, fog and snow),
- light conditions
- speed.

This therefore means that a particular trail, on particular days, could also be much more difficult to deal with than under ideal conditions, such as sunny weather and dry terrain.

In the STS Single Trail Scale the following criteria were taken into account when assigning grades:

- Constitution of the trail, i.e. the grip and type of surface
- Types of obstacles
- Gradient
- Category of curves
- Required riding skills.

The average difficulty level S of a route in the Single Trail Scale

In the real world, the same single track can present different levels of difficulty in some of its passages or sections. As convention, what determines the classification of a single track in the STS mtb trail difficulty rating system is the average difficulty level S detected for most of the route. A route can therefore be classified as S2 even if some of its passages and sections are rated S3.

The 5 Levels of the STS Mtb Trail Difficulty Rating System

S0 indicates a single trail that does not present particular difficulties.
Most of them are forest or meadow trails with good grip, or compact gravel. In this type of trail you will not face steps, rocks or roots. The gradient of the trail goes from mild to moderate and the curves are always quite wide. S0 trails can be tackled without special mountain bike technical skills.

- **TRAIL CONDITIONS: compact terrain with good grip**
- **OBSTACLES: none**
- **GRADIENT: mild to moderate**
- **CURVES: wide**
- **MTB TECH SKILLS: no special skills required**

+ We include the double track with rides on hard-pack gravel/dirt

The doubletrack is relatively wide where you can cycle side by side and offer a more social riding experience, perfect for enjoying the scenery together

On an S1 trail you will find small obstacles such as not very prominent roots and small stones.

Often the increased difficulty rating is due to the drains or erosion. In S1 single trails the surface could be not very compact. The gradient reaches a maximum of 40% and no switchbacks will be encountered. Starting from S1 level, basic mtb riding techniques and good concentration are required. The trickier passages require the ability to dose the brakes and to affect the trajectory by body displacement. Obstacles can all be rolled over with basic riding skills.

- **TRAIL CONDITIONS: loose surface possible, with small roots and stones**
- **OBSTACLES: small obstacles (drains, damage caused by erosion)**
- **GRADIENT: <40%**
- **CURVES: narrow**
- **MTB TECH SKILLS: basic driving skills required**

Within the S2 classification you will find trails with larger roots, stones, steps and easy staircases.

Often there are narrow curves and gradients can reach 70% in some passages or sections. A fair amount of mtb riding ability is required to ridden obstacles.

The ability to brake at all times and to shift your center of gravity are necessary techniques, as well as the ability to precisely control the braking and to keep the body always active while driving.

- **TRAIL CONDITIONS: surface most often lose, protruding roots and stones**
- **OBSTACLES: obstacles of various kinds and staircases**
- **GRADIENT: <70%**
- **CURVES: easy switchbacks**
- **MTB TECH SKILLS: advanced driving skills required**

Blocked single trails with many larger boulders and / or root passages belong to category S3.

There are often high steps, switchbacks and tricky traverses. You will have to deal also with slippery and not very compact terrain. Sections with a gradient over 70% are not rare. S3 sections do not yet require trial technique, however excellent control of the bike, mastery in the brake modulation and an excellent balance are required.

- **TRAIL CONDITIONS: technical, frequent protruding roots and large rocks, slippery ground and not very compact surface**
- **OBSTACLES: high boulders**
- **GRADIENT: >70%**
- **CURVES: tight switchbacks**
- **MTB TECH SKILLS: more than advanced driving skills required**

S4 describes very steep and blocked single trails with large blocks of rocks and / or challenging root passages, as well as loose surface.

Extremely steep slopes, narrow switchbacks and steps high enough to get in contact with the crown of the crank are common. To properly ride a S4 trail, trial techniques such as the ability to shift the front wheel and rear wheel, a perfect braking technique and an excellent balance are absolutely necessary. Only extreme bikers can tackle an **S4 route. Even carrying a bike up such passages is often quite difficult.**

- **TRAIL CONDITIONS: technical, frequent protruding roots and large rocks, slippery and not compact terrain**
- **OBSTACLES: steep ramps, very high steps (often at crown height or more)**
- **GRADIENT: >70%**
- **CURVES: very tight switchbacks**
- **MTB TECH SKILLS: perfect mastery of the bike and trial skills such as displacement of the rear wheel in sharp turns.**

The S5 grade is characterized by a very technical terrain with counterslopes and slippery surface, extremely tight curves, extreme steepness (cliffs), very big obstacles in close sequence including fallen trees. Braking distance is very short if any. Some obstacles can only be jumped over. Hairpins are so tight that shifting the wheels is hardly possible, even carrying the bike would be almost impossible, as you need your hands to hold on or even climb. Only a handful of freaks tackle S5 passages.

- **TRAIL CONDITIONS: very technical with counterslopes, slippery and not compact terrain. Can include sections similar to high mountain climbing routes.**
- **OBSTACLES: steep ramps, steps in close sequence difficult to overcome**
- **GRADIENT: >>70%**
- **CURVES: very tight bends with obstacles**
- **MTB TECH SKILLS: excellent mastery of trial techniques; shifting the front and rear wheel is very limited.**

(*) The Single Trail Skala (or Single Trail Scale – STS) has been developed by Carsten Schymik, David Werner and Harald Philipp.

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